Brownian regularity of the KPZ fixed point

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Rayleigh-Knight Prize

Joint work with Sourav Sarkar

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Overview

1. The KPZ universality class
2. The KPZ fixed point
3. Brownianness of the KPZ fixed point

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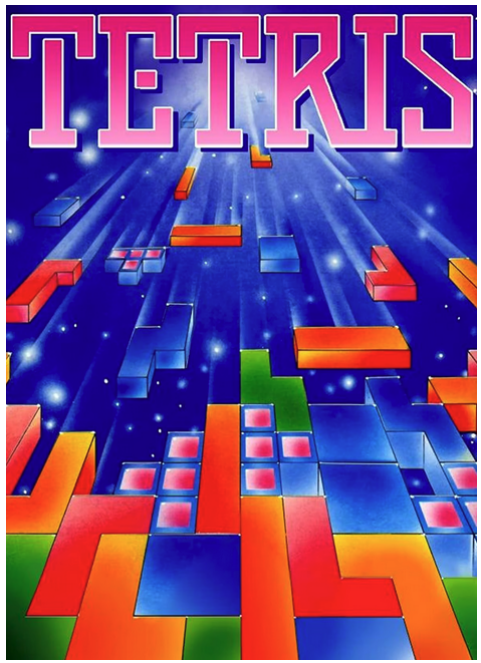
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Growing interfaces in the real world







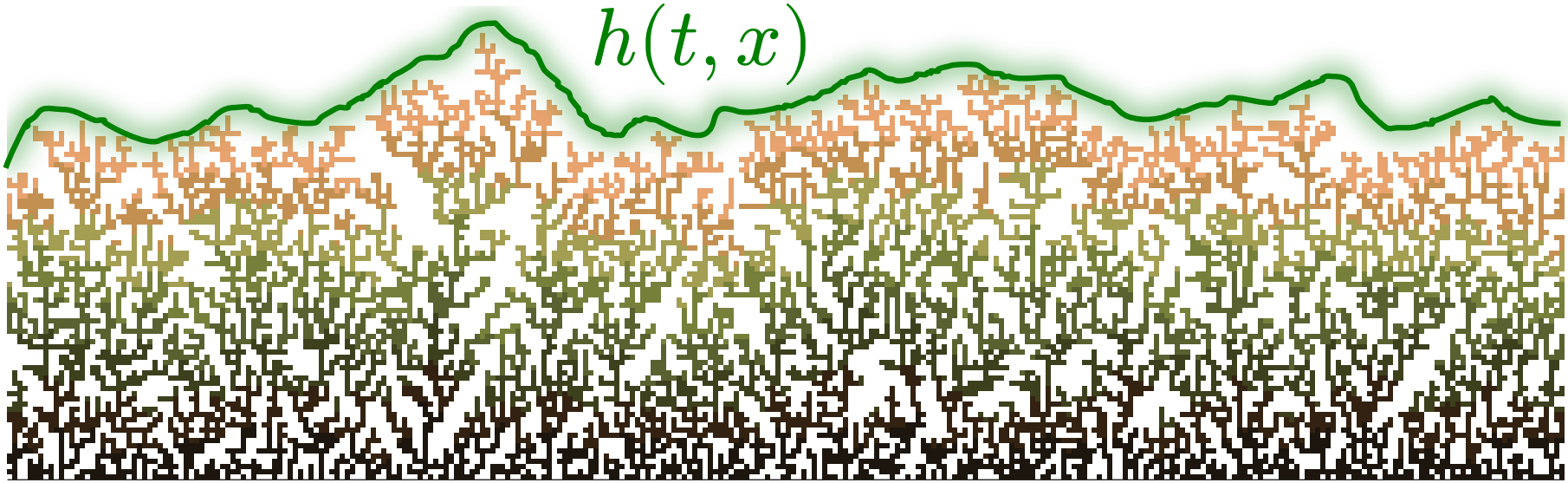


What is the KPZ universality class?

- Kardar-Parisi-Zhang (1986) predicted that many planar random growth processes possess universal scaling behaviour, with local dynamics governed by the KPZ equation:

$$\partial_t h = \underbrace{(\partial_x h)^2}_{\text{lateral growth}} + \underbrace{\partial_x^2 h}_{\text{smoothing}} + \underbrace{\xi}_{\text{spacetime white noise}}$$

$$\underbrace{h(t,x)}_{\text{height at time } t\text{, position } x}$$



What is the KPZ universality class?

A model is in the KPZ universality class if it has an associated height function that converges under the $1:2:3$ scaling:

$$h(t,x) \mapsto \varepsilon h(\varepsilon^{-3}t, \varepsilon^{-2}x) - C_\varepsilon \cdot t, \text{ as } \varepsilon \searrow 0$$

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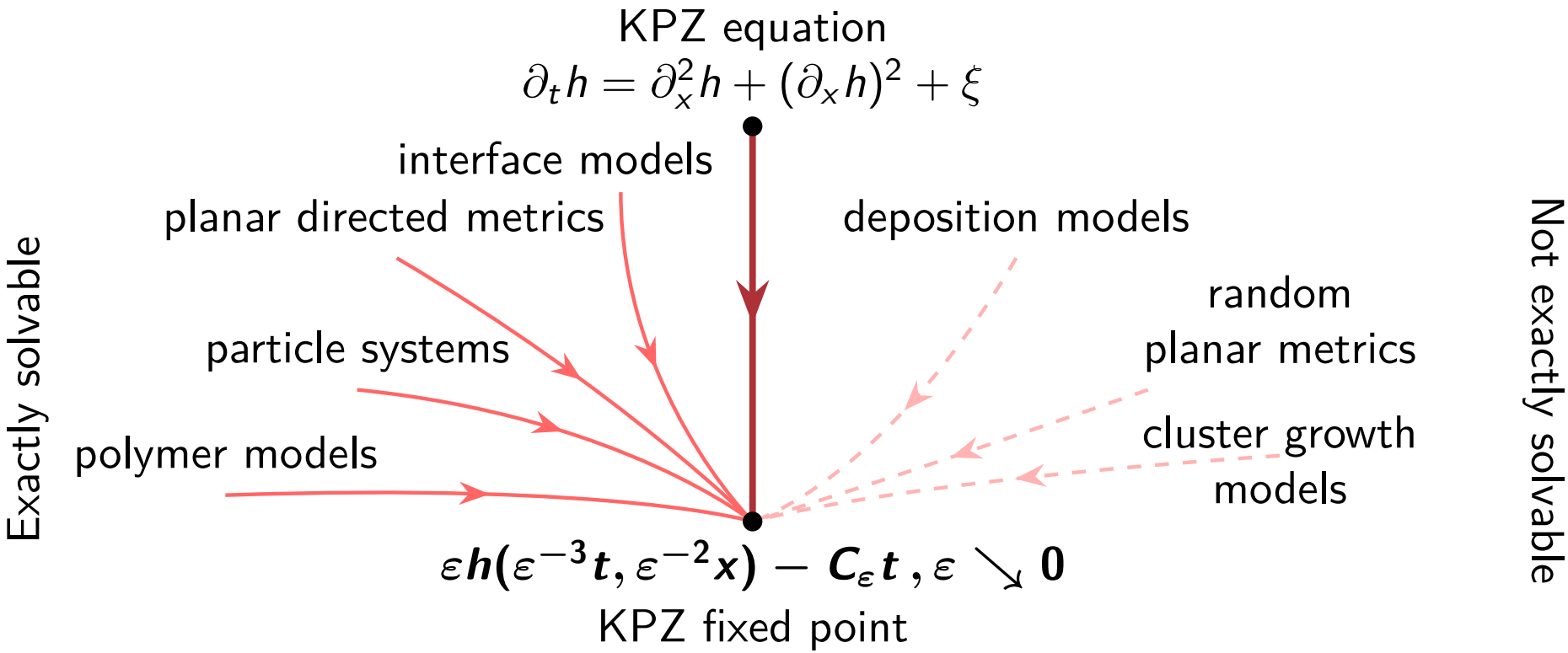
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- It can be thought of as a random evolution:

$$\underbrace{h_0}_{\text{initial profile}} \longrightarrow \underbrace{\mathfrak{h}_t}_{\text{interface at time } t} \longrightarrow \underbrace{\mathfrak{h}_{t+s}}_{\text{interface at time } t+s} \longrightarrow \cdots$$

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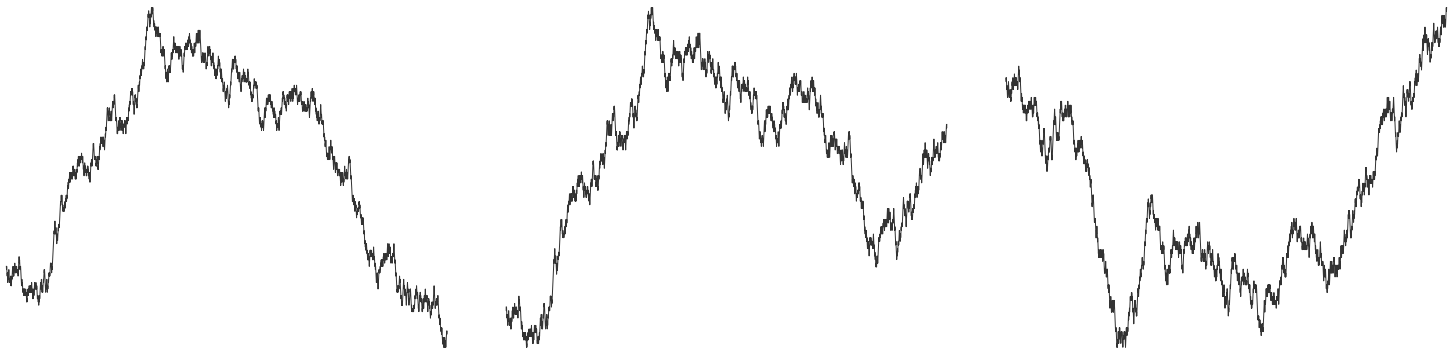


Figure: The KPZ fixed point (at unit time) started from various initial height profiles.

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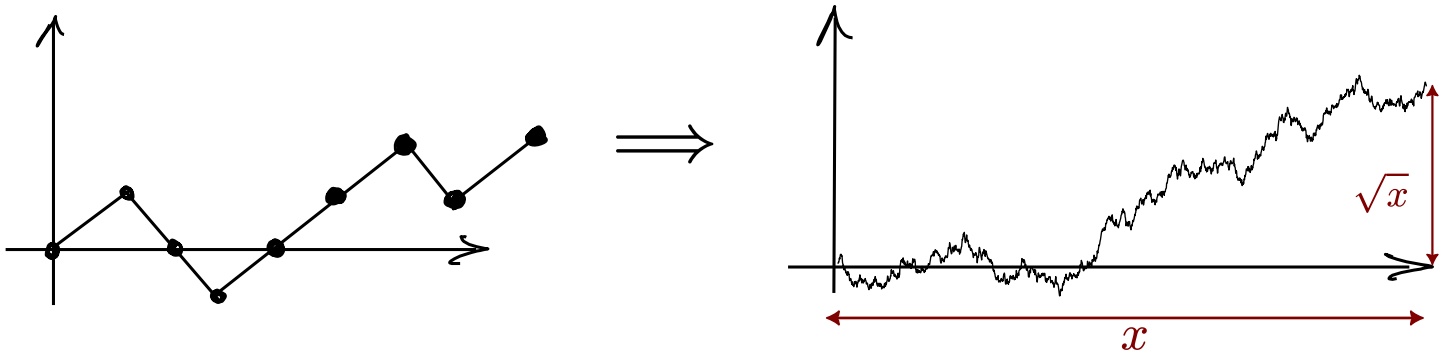
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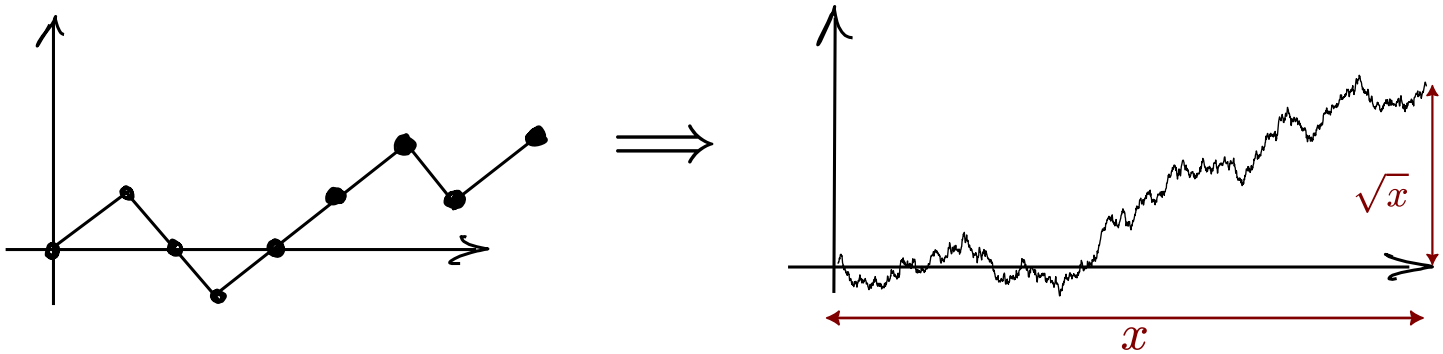
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- Brownian motion is invariant and attractive for the KPZ fixed point.

Long quest

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- Corwin-Hammond (2014): Brownian on compacts for Airy_2
- Pimentel (2018): local Brownian limit and Hölder continuity for certain initial conditions
- M-Q-R (2016), Pimentel (2019): Hölder $1/2-$ continuity and local limit convergence for KPZ fixed point
- Calvert-Hegde-H (2019): For general initial conditions – Brownian patchwork quilt. Question left open: if it can be reduced to a single patch?
- Sarkar-Virág (2021): For all initial conditions, the KPZ fixed point at any time t is Brownian on compacts.
- Dauvergne (2024): the Airy_2 process has a bounded density with respect to Brownian motion.

Main results

- From S-V (2021), with ν_{KPZ} the law of the KPZ fixed point started from some arbitrary initial profile and μ that of Brownian motion on a bounded interval, we can express for any event A:

$$\nu_{\mathrm{KPZ}}(A) = \int_A X \mathrm{d}\mu$$

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- Our main result is a quantitative absolute continuity relation between ν_{KPZ} and μ: for an event A,

$$\mu(A) \leq \delta \implies \nu_{\mathrm{KPZ}}(A) \leq f(\delta)\,,$$

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Theorem (Sarkar-T. (2025))

For all $p > 1$, $\int X \cdot (|\log X|)^p \mathrm{d}\mu$ is finite.

Future directions

- Conjecture in C-H-H (2019): can we obtain for all $p > 1$, $\int (X)^p \mathrm{d}\mu$ is finite?

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- Can we distinguish the KPZ fixed points started from different initial data?

Thanks!

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